

Pearson Edexcel International GCSE

(9-1) ICT: Web Authoring Masterclass

Web authoring software is used to design, create, and edit web pages and complete websites. Web pages are essentially documents written in **HTML (Hypertext Markup Language)**, which is the universal structural language of the World Wide Web.

Web browser software (such as Google Chrome, Microsoft Edge, or Mozilla Firefox) reads the HTML code, interprets the instructions or "markup tags," and displays the resulting visual page. To author websites, designers either write raw code in a plain-text editor or use specialized **WYSIWYG** (What You See Is What You Get) applications like Adobe Dreamweaver, KompoZer, or Amaya, which generate the underlying HTML automatically.

1. Website Infrastructure: Templates, Navigation, and Layouts

A professional website must maintain visual consistency and clear usability across all its pages. This is achieved using structural layout templates.

The Purpose of Templates

A **template** is a master page layout template used to create multiple web pages with a matching look, feel, and structure. Templates save development time and ensure that branding elements remain identical across the entire site.

Standard Page Features

Web pages are typically broken down into distinct structural zones using HTML layout blocks or tables:

- **Banner / Header:** Positioned at the very top of the page canvas. It typically contains the company logo, the website title, and primary corporate branding graphics.
- **Navigation Menu:** A structured set of buttons or text hyperlinks clustered together (either horizontally below the banner or vertically down the left sidebar). The navigation menu must be clear, intuitive, and appear in the exact same position on every single page so users can navigate the site easily.
- **Sidebar:** A narrow vertical column used to host secondary links, search bars, or social media widgets.
- **Main Body Content:** The primary space reserved for the unique text, images, and videos of that specific web page.
- **Footer:** Positioned at the absolute bottom of the page. It contains copyright notices, privacy policy links, and contact information.

2. Page Content & Asset Integration

Building an engaging web page requires integrating different digital assets directly into the layout.

- **Text Formatting:** Web text is structured hierarchically using strict formatting styles to make the content easy to scan:
 - **Headings (<h1>):** Large, prominent text used for the main page title.
 - **Sub-headings (<h2>, <h3>):** Medium text used to break content into logical sub-sections.
 - **Body Text (<p>):** Standard scale text optimized for reading paragraphs.
- **Images:** Graphic files (such as .png or .jpeg) inserted to support the text. Images require an **Alt Text (Alternative Text)** description embedded in the code. Alt text is read aloud by screen readers for visually impaired visitors and displays on screen if the image file fails to load.
- **Buttons & Interactivity:** Interactive graphic blocks or text elements designed to trigger actions when clicked, such as submitting a form or jumping to another section.
- **Animations:** Embedded moving elements (such as .gif loops or HTML5 assets) used to draw attention to specific announcements or banners.

3. Mastering Core HTML Code & Tags

HTML uses **tags** enclosed in angle brackets (< and >) to instruct web browsers how to handle and display content. Most markup tags work in pairs: an opening tag to start an effect and a closing tag containing a forward slash (/) to end it.

Essential Text Enhancement Tags

- **:** Instructs the browser to display text in **bold** to signify strong importance.
 - *Syntax:* Bold Text
- **:** Applies *emphasis* to text by rendering it in italics.
 - *Syntax:* Italicized Text
- **<u>:** Displays text with a structural <u>underline</u>.
 - *Syntax:* <u>Underlined Text</u>

Structural Layout Tags

- **<p>:** Defines a standard paragraph. The browser automatically injects a wider vertical space buffer between paragraph blocks to ensure readability.
- **
:** Inserts a direct **line break**. Unlike paragraph tags, a line break forces text to drop down to the next line immediately *without* adding extra vertical spacing. **Note:**
 is a standalone or "empty" tag; it does not require a closing </br> tag.

Hyperlinks and Asset Insertion

- **Inserting Images ():** Links an external image file into the page layout. It utilizes key attributes to specify the file source location, alt text, and dimensions:
 - *Syntax:*
- **Inserting Hyperlinks (<a>):** Creates an active link connecting the page to another asset or external website using the href (Hypertext Reference) attribute.

- *Syntax:* `<a href="http://www.pearson.com" target="_blank">Visit Pearson</a>`
- *The Target Attribute:* Adding `target="_blank"` instructs the web browser to open the clicked link in a completely new window or tab, preventing the user from navigating away from your original website.

Student Activities

Activity 1: Visualizing Page Structures

1. Open any major educational or news website in your web browser.
2. Carefully sketch the structural layout framework of the homepage on a sheet of paper.
3. Label the boundaries of the **Banner**, the **Navigation Menu**, the **Main Body Content**, and the **Footer**.
4. Navigate to a different sub-page on the same website. Note down whether the banner and navigation menu remain in the exact same positions, and explain how this consistency benefits the user experience.

Activity 2: Writing and Interpreting Raw HTML

1. Open a basic plain-text editor (such as *Notepad* on Windows or *TextEdit* on Mac configured to plain text mode).
2. Carefully type out the structural HTML template code block displayed below:

```
<!DOCTYPE html>
<html>
<head>
  <title>My First ICT Web Page</title>
</head>
<body>
  <h1>Welcome to Web Authoring</h1>
  <p>This is a standard body paragraph containing <strong>important
bold words</strong>.</p>
  <p>This line ends here<br>and drops down immediately using a line
break tag.</p>
  <p><a href="https://www.wikipedia.org" target="_blank">Click here
to visit Wikipedia in a new tab</a></p>
</body>
</html>
```

1. Save the file with the exact name `index.html` (ensuring the extension is `.html` and not `.txt`).
2. Locate the saved file in your file explorer and double-click it to open it in a web browser. Verify that the font enhancements, line breaks, and hyperlinks render and function correctly.

Chapter Questions

1. Which HTML tag is used to force text to drop down to the very next line instantly without adding a wide paragraph space buffer? A) `<p>` B) `<href>` C) `<br>` D) `<strong>`

2. Explain why web designers utilize layout templates when building large websites containing multiple pages.
3. Describe the role of a web browser application when interacting with an HTML file.
4. Which attribute must be included inside an <a> anchor tag to specify the exact web address destination of a hyperlink? A) src B) alt C) target D) href
5. State the purpose of embedding Alt Text (alt) descriptions within an image tag structure.
6. Explain the structural difference on screen between wrapping text inside paragraph tags (<p>...</p>) versus separating text lines using a line break tag (
).
7. State the exact HTML code snippet required to display the text "Revision Master" in bold styling.
8. State the purpose of adding the target="_blank" attribute configuration into a web page hyperlink anchor tag.
9. Identify two standard design features that are typically locked into a website's master header banner zone.
10. Which character is used inside an HTML tag to signify to the web browser that a specific markup effect is ending? A) ? B) ! C) / D) *

Answer Key & Explanations

1. **Correct Answer: C (
)** *Explanation:* The
 tag executes a clean line break, dropping text down to the immediate next line while maintaining tight line spacing.
2. **Core Reasons:** * Templates ensure visual consistency across the entire website, making it look professional.
 - They guarantee that critical navigation elements stay in identical positions on every page, preventing user confusion.
 - They save development time because designers do not have to rebuild the header, footer, and navigation layouts from scratch for every new page.
3. **Core Browser Description:** * A web browser reads and processes the raw instructions written in HTML code.
 - It translates the markup tags into structured layout visuals, paragraphs, and active hyperlinks, rendering the final human-readable web page on screen.
4. **Correct Answer: D (href)** *Explanation:* The href (Hypertext Reference) attribute holds the exact URL path or file destination target that the hyperlink points to.
5. **Core Alt Text Purposes:** * It assists accessibility by providing text descriptions that screen readers can read aloud to visually impaired site visitors.
 - It acts as a fallback display asset, showing descriptive text on screen if the website loses connection to the image file and fails to load it.
6. **Core Structural Difference:** * Paragraph tags (<p>) create a distinct block of text and automatically inject a wider vertical blank space before and after the block.
 - A line break (
) drops the text down immediately on a single line within the same block, keeping the line spacing narrow.
7. **Correct Code Snippet:** * Revision Master (or Revision Master)
8. **Core Attribute Purpose:** * It forces the web browser to launch the hyperlinked address in a completely separate browser tab or window, keeping the user's current session on the original web page active.
9. **Standard Banner Features:** * Corporate or organizational logos.

- Main website title headings / branding graphics.

10. Correct Answer: C (I) *Explanation:* The forward slash character placed directly before the tag name indicates a closing HTML tag (e.g., closing a paragraph with </p>), telling the browser to stop applying that specific formatting rule.